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Instructions:

1. Write a program to:

- Prompt the user for input for a score between 0.0 and 1.0.
- If the score is out of range, print an error.
- If the score is between 0.0 and 1.0, print a grade using the following:
A: ≥ 0.9
B: ≥ 0.8
C: ≥ 0.7
D: ≥ 0.6
F: < 0.6
- If the user enters a value out of range, print a suitable error message. (Only need to get one input, no need to loop for more inputs.)
- **For the test**, enter a score of 0.85 and you should get a B.

2. Write a program to:

- Prompt the user for hours and rate per hour.
- Compute base pay for rate * hours up to 40 hours.
- Award time-and-a-half (1.5) for the hourly rate for all hours worked above 40 hours.
- The program should return a value of total pay (base pay + overtime pay).
- **Test data:** Use 45 hours and a rate of 10.50 per hour to test the program (the pay should be 498.75).
- Do not worry about error checking the user input unless you want to - you can assume the user types numbers properly.

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In [1]: # Part1
def get_score_grade(score):
    if score >= 0.9:
        score_grade = 'A'
    elif score >= 0.8:
        score_grade = 'B'
    elif score >= 0.7:
        score_grade = 'C'
    elif score >= 0.6:
        score_grade = 'D'
    else:
        score_grade = 'F'
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    return score_grade

# Prompt for input
input_score = input("Enter a score between 0.0 and 1.0 >> ")
try:
    # Try to transform score to float
    score = float(input_score)

    # Range check
    if score < 0.0 or score > 1.0:
        print("Error: Score is out of range!")
    else:
        # Print the input data and final grade
        print("-" * 15)
        print(f"Score = {score:.2f}")
        print("-" * 15)
        print(f"Score Grade: {get_score_grade(score)}")
except ValueError:
    print("Error: Please enter numeric input!")

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Score = 0.85
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Score Grade: B

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In [2]: # Part2

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BASE_HOURS = 40          # Limit on working hours at the regular rate
OT_RATE = 1.5            # Overtime pay rate (one and a half times the regular rate)

try:
    # Prompt for input and transform to float
    hours_worked = float(input("Enter hours:"))
    hourly_rate = float(input("Enter rate per hour:"))

    # Calculate gross pay: regular rate for up to 40 hours, 1.5x for overtime
    if hours_worked <= 40:
        gross_pay = hours_worked * hourly_rate
    else:
        overtime_hours = hours_worked - BASE_HOURS
        gross_pay = (BASE_HOURS * hourly_rate) + (overtime_hours * hourly_rate * OT_RATE)

    # Print the Input data and Gross pay
    print("-" * 22)
    print(f"Hours worked = {hours_worked:.0f} \nRate per hour = {hourly_rate:.2f}")
    print("-" * 22)
    print(f"Gross pay = {gross_pay:.2f}")

except ValueError:
    print("Error: Please enter numeric input!")

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Hours worked = 45

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Rate per hour = 10.50
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Gross pay = 498.75

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In []: